

# SIEM Integration with Splunk

## Objective :

To integrate Splunk (a SIEM platform) into a simulated enterprise environment for centralized monitoring, log collection, and incident detection. The setup enhances visibility into system activities, identifies brute-force attempts, and automates blocking using UFW firewall rules.

## Scope :

- Installation and configuration of Splunk Enterprise on the server.
- Installation and setup of Splunk Universal Forwarder on the Ubuntu client.
- Log Sources Integrated:
  - Authentication Logs
  - Firewall Logs
  - Web Server Logs (Apache)
- Deployment and configuration of UFW to mitigate brute-force login attacks.
- Log integration into Splunk for centralized monitoring.
- Creation of dashboards and alerts in Splunk for:
  - Failed login attempts.
  - Blocked IP addresses by UFW
  - UFW firewall activity.

## Implementation Steps :

Splunk Installation & Setup :

- Installed Splunk Enterprise on the Linux sever.

```

(root@kali)-[/opt]
# /opt/splunk/bin/splunk start --accept-license
systemctl: /opt/splunk/lib/libcrypto.so.3: version 'OPENSSL_3.4.0' not found (required by /usr/lib/x86_64-linux-gnu/systemd/libsystemd-shared-257.so)

This appears to be your first time running this version of Splunk.

Splunk software must create an administrator account during startup. Otherwise, you cannot log in.
Create credentials for the administrator account.
Characters do not appear on the screen when you type in credentials.

Please enter an administrator username: saniyya
Password must contain at least:
  * 8 total printable ASCII character(s).
Please enter a new password:
Please confirm new password:
Copying '/opt/splunk/etc/openldap/ldap.conf.default' to '/opt/splunk/etc/openldap/ldap.conf'.
Writing RSA key
Writing RSA key
Moving '/opt/splunk/share/splunk/search_mrsparkle/modules.new' to '/opt/splunk/share/splunk/search_mrsparkle/modules'.
systemctl: /opt/splunk/lib/libcrypto.so.3: version 'OPENSSL_3.4.0' not found (required by /usr/lib/x86_64-linux-gnu/systemd/libsystemd-shared-257.so)

Splunk> 4TW

Checking prerequisites...
  Checking http port [8000]: open
  Checking mgmt port [8089]: open
  Checking appserver port [127.0.0.1:8085]: open
  Checking kvstore port [8191]: open
  Checking configuration... Done.
    Creating: /opt/splunk/var/lib/splunk
    Creating: /opt/splunk/var/run/splunk
    Creating: /opt/splunk/var/run/splunk/appserver/i18n
    Creating: /opt/splunk/var/run/splunk/appserver/modules/static/css
    Creating: /opt/splunk/var/run/splunk/upload
    Creating: /opt/splunk/var/run/splunk/search_telemetry
    Creating: /opt/splunk/var/run/splunk/search_log
    Creating: /opt/splunk/var/spool/splunk
    Creating: /opt/splunk/var/spool/dirmoncache
    Creating: /opt/splunk/var/lib/splunk/authDb
    Creating: /opt/splunk/var/lib/splunk/hashDb
    Creating: /opt/splunk/var/run/splunk/collect
    Creating: /opt/splunk/var/run/splunk/sessions
New certs have been generated in '/opt/splunk/etc/auth'.
New certs have been generated in '/opt/splunk/etc/auth'.
  Checking critical directories... Done
  Checking indexes...
    Validated: _audit _configtracker _dsappevent _dsclient _dsphonehome _internal _introspection _metrics _metrics_rollup _telemetry _thefishbucket history main summary
  Done
  Checking filesystem compatibility... Done
  Checking conf files for problems...

```

- Installed Splunk Universal Forwarder on the Ubuntu target system.

```

root@ubuntu:/opt# /opt/splunkforwarder/bin/splunk start --accept-license
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"

This appears to be your first time running this version of Splunk.

Splunk software must create an administrator account during startup. Otherwise, you cannot log in.
Create credentials for the administrator account.
Characters do not appear on the screen when you type in credentials.

Please enter an administrator username: saniyya
Password must contain at least:
  * 8 total printable ASCII character(s).
Please enter a new password:
Please confirm new password:
Creating unit file...
Important: splunk will start under systemd as user: splunkfwd
The unit file has been created.

Splunk> 4TW

Checking prerequisites...
  Checking mgmt port [8089]: open
    Creating: /opt/splunkforwarder/var/lib/splunk
    Creating: /opt/splunkforwarder/var/run/splunk
    Creating: /opt/splunkforwarder/var/run/splunk/appserver/i18n
    Creating: /opt/splunkforwarder/var/run/splunk/appserver/modules/static/css
    Creating: /opt/splunkforwarder/var/run/splunk/upload
    Creating: /opt/splunkforwarder/var/run/splunk/search_telemetry
    Creating: /opt/splunkforwarder/var/run/splunk/search_log
    Creating: /opt/splunkforwarder/var/spool/splunk
    Creating: /opt/splunkforwarder/var/spool/dirmoncache
    Creating: /opt/splunkforwarder/var/lib/splunk/authDb
    Creating: /opt/splunkforwarder/var/lib/splunk/hashDb
    Creating: /opt/splunkforwarder/var/run/splunk/collect
    Creating: /opt/splunkforwarder/var/run/splunk/sessions
New certs have been generated in '/opt/splunkforwarder/etc/auth'.
New certs have been generated in '/opt/splunkforwarder/etc/auth'.
  Checking conf files for problems...

```

- Configured the forwarder to send system logs (`/var/log`) to the Splunk server.

```

root@laalu-VirtualBox:~# /opt/splunkforwarder/bin/splunk add monitor /var/log
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"
Your session is invalid. Please login.
Splunk username: laalu
Password:
Cannot create another input with the name "/var/log", one already exists.
root@laalu-VirtualBox:~#

```

## Splunk Dashboards & Alerts :

- Brute-force attack attempts
  - System-level (SSH):** Multiple failed login attempts captured from `/var/log/auth.log` on the Ubuntu target machine.

| < Hide Fields      | All Fields | Time    | Event                                                                                                                       |
|--------------------|------------|---------|-----------------------------------------------------------------------------------------------------------------------------|
| SELECTED FIELDS    |            | 8/20/25 | 2025-08-20T20:44:05.37650+05:30 laalu-VirtualBox sshd[3729]: Failed password for laalu from 192.168.206.112 port 34446 ssh2 |
| host 1             |            | 8/20/25 | 2025-08-20T20:44:05.37650+05:30 laalu-VirtualBox sshd[3729]: Failed password for laalu from 192.168.206.112 port 34446 ssh2 |
| source 3           |            | 8/20/25 | 2025-08-20T20:44:05.37650+05:30 laalu-VirtualBox sshd[3729]: Failed password for laalu from 192.168.206.112 port 34446 ssh2 |
| source_type 3      |            | 8/20/25 | 2025-08-20T20:44:05.37650+05:30 laalu-VirtualBox sshd[3729]: Failed password for laalu from 192.168.206.112 port 34446 ssh2 |
| INTERESTING FIELDS |            | 8/20/25 | 2025-08-20T20:44:05.37650+05:30 laalu-VirtualBox sshd[3729]: Failed password for laalu from 192.168.206.112 port 34446 ssh2 |
| date_hour 1        |            | 8/20/25 | 2025-08-20T20:44:05.37650+05:30 laalu-VirtualBox sshd[3729]: Failed password for laalu from 192.168.206.112 port 34446 ssh2 |
| date_minute 7      |            | 8/20/25 | 2025-08-20T20:44:05.37650+05:30 laalu-VirtualBox sshd[3729]: Failed password for laalu from 192.168.206.112 port 34446 ssh2 |
| date_second 26     |            | 8/20/25 | 2025-08-20T20:44:05.37650+05:30 laalu-VirtualBox sshd[3729]: Failed password for laalu from 192.168.206.112 port 34446 ssh2 |
| date_year 1        |            | 8/20/25 | 2025-08-20T20:44:05.37650+05:30 laalu-VirtualBox sshd[3729]: Failed password for laalu from 192.168.206.112 port 34446 ssh2 |
| date_zone 1        |            | 8/20/25 | 2025-08-20T20:44:05.37650+05:30 laalu-VirtualBox sshd[3729]: Failed password for laalu from 192.168.206.112 port 34446 ssh2 |
| index 1            |            | 8/20/25 | 2025-08-20T20:44:05.37650+05:30 laalu-VirtualBox sshd[3729]: Failed password for laalu from 192.168.206.112 port 34446 ssh2 |
| punct 52           |            | 8/20/25 | 2025-08-20T20:44:05.37650+05:30 laalu-VirtualBox sshd[3729]: Failed password for laalu from 192.168.206.112 port 34446 ssh2 |
| splunk_server 1    |            | 8/20/25 | 2025-08-20T20:44:05.37650+05:30 laalu-VirtualBox sshd[3729]: Failed password for laalu from 192.168.206.112 port 34446 ssh2 |
| timestamp 2        |            | 8/20/25 | 2025-08-20T20:44:05.37650+05:30 laalu-VirtualBox sshd[3729]: Failed password for laalu from 192.168.206.112 port 34446 ssh2 |
| timestamp 2        |            | 8/20/25 | 2025-08-20T20:44:05.37650+05:30 laalu-VirtualBox sshd[3729]: Failed password for laalu from 192.168.206.112 port 34446 ssh2 |
| 30 more fields     |            | 8/20/25 | 2025-08-20T20:44:05.37650+05:30 laalu-VirtualBox sshd[3729]: Failed password for laalu from 192.168.206.112 port 34446 ssh2 |
| Extract New Fields |            | 8/20/25 | 2025-08-20T20:44:05.37650+05:30 laalu-VirtualBox sshd[3729]: Failed password for laalu from 192.168.206.112 port 34446 ssh2 |

- **Server-level (DVWA Web Application)** :Repeated failed login attempts on the DVWA web interface, captured through Apache web logs.

| < Hide Fields      | All Fields | Time | Event |
|--------------------|------------|------|-------|
| SELECTED FIELDS    |            |      |       |
| host 1             |            |      |       |
| source 5           |            |      |       |
| sourcetype 4       |            |      |       |
| INTERESTING FIELDS |            |      |       |
| bytes 16           |            |      |       |
| clreng 2           |            |      |       |
| date_hour 1        |            |      |       |
| date_mday 1        |            |      |       |
| date_minute 7      |            |      |       |
| date_month 1       |            |      |       |
| date_offset 43     |            |      |       |
| date_week 1        |            |      |       |
| date_year 1        |            |      |       |
| date_zone 2        |            |      |       |
| file 8             |            |      |       |
| index 1            |            |      |       |
| index 1            |            |      |       |
| logid 1            |            |      |       |
| method 100+        |            |      |       |
| punct 100+         |            |      |       |
| referer 35         |            |      |       |
| req_time 35        |            |      |       |

|   |                |                                        |                                                                                                                          |
|---|----------------|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| > | 8/21/25        | 192.168.206.112                        | - - [21/Aug/2025:07:37:15 +0530] "GET /DVWA/vulnerabilities/brute/?username=admin&password=fairoosDarwin" 302 0 "-" "-"  |
|   | 7:37:15.000 AM | host = <a href="#">kahu-VirtualBox</a> | source = /var/log/apache2/access.log : sourcetype = access_combined                                                      |
| > | 8/21/25        | 192.168.206.112                        | - - [21/Aug/2025:07:37:15 +0530] "GET /DVWA/vulnerabilities/brute/?username=admin&password=fairoosDaniel" 302 0 "-" "-"  |
|   | 7:37:15.000 AM | host = <a href="#">kahu-VirtualBox</a> | source = /var/log/apache2/access.log : sourcetype = access_combined                                                      |
| > | 8/21/25        | 192.168.206.112                        | - - [21/Aug/2025:07:37:15 +0530] "GET /DVWA/vulnerabilities/brute/?username=admin&password=fairoosDakota" 302 0 "-" "-"  |
|   | 7:37:15.000 AM | host = <a href="#">kahu-VirtualBox</a> | source = /var/log/apache2/access.log : sourcetype = access_combined                                                      |
| > | 8/21/25        | 192.168.206.112                        | - - [21/Aug/2025:07:37:15 +0530] "GET /DVWA/vulnerabilities/brute/?username=admin&password=fairoosDallas" 302 0 "-" "-"  |
|   | 7:37:15.000 AM | host = <a href="#">kahu-VirtualBox</a> | source = /var/log/apache2/access.log : sourcetype = access_combined                                                      |
| > | 8/21/25        | 192.168.206.112                        | - - [21/Aug/2025:07:37:15 +0530] "GET /DVWA/vulnerabilities/brute/?username=admin&password=fairoosDawau" 302 0 "-" "-"   |
|   | 7:37:15.000 AM | host = <a href="#">kahu-VirtualBox</a> | source = /var/log/apache2/access.log : sourcetype = access_combined                                                      |
| > | 8/21/25        | 192.168.206.112                        | - - [21/Aug/2025:07:37:15 +0530] "GET /DVWA/vulnerabilities/brute/?username=admin&password=fairoosDIGITAL" 302 0 "-" "-" |
|   | 7:37:15.000 AM | host = <a href="#">kahu-VirtualBox</a> | source = /var/log/apache2/access.log : sourcetype = access_combined                                                      |
| > | 8/21/25        | 192.168.206.112                        | - - [21/Aug/2025:07:37:15 +0530] "GET /DVWA/vulnerabilities/brute/?username=admin&password=fairoosDRAGON" 302 0 "-" "-"  |
|   | 7:37:15.000 AM | host = <a href="#">kahu-VirtualBox</a> | source = /var/log/apache2/access.log : sourcetype = access_combined                                                      |
| > | 8/21/25        | 192.168.206.112                        | - - [21/Aug/2025:07:37:15 +0530] "GET /DVWA/vulnerabilities/brute/?username=admin&password=fairoosDEMO" 302 0 "-" "-"    |
|   | 7:37:15.000 AM | host = <a href="#">kahu-VirtualBox</a> | source = /var/log/apache2/access.log : sourcetype = access_combined                                                      |
| > | 8/21/25        | 192.168.206.112                        | - - [21/Aug/2025:07:37:15 +0530] "GET /DVWA/vulnerabilities/brute/?username=admin&password=fairoosDEMO5" 302 0 "-" "-"   |
|   | 7:37:15.000 AM | host = <a href="#">kahu-VirtualBox</a> | source = /var/log/apache2/access.log : sourcetype = access_combined                                                      |

## ● Dashboard and alerts

127.0.0.1:8000/en-US/app/search/alerts?%2Fservices%2Ffairoos%2Fsearch%2Fsaved%2Fsearches%2FBrute%2520Force%2520and%2520...

spunk-enterprise Apps

Administrator Messages Settings Activity Help Find

Search Analytics Datasets Reports Alerts Dashboards Search & Reporting

### Brute Force and Web Attack Alert

Enabled: Yes, Disable

App: Search

Permissions: Private, Owned by fairoos, Edit

Modified: Aug 21, 2025 7:47:04 AM

Alert Type: Real time, Edit

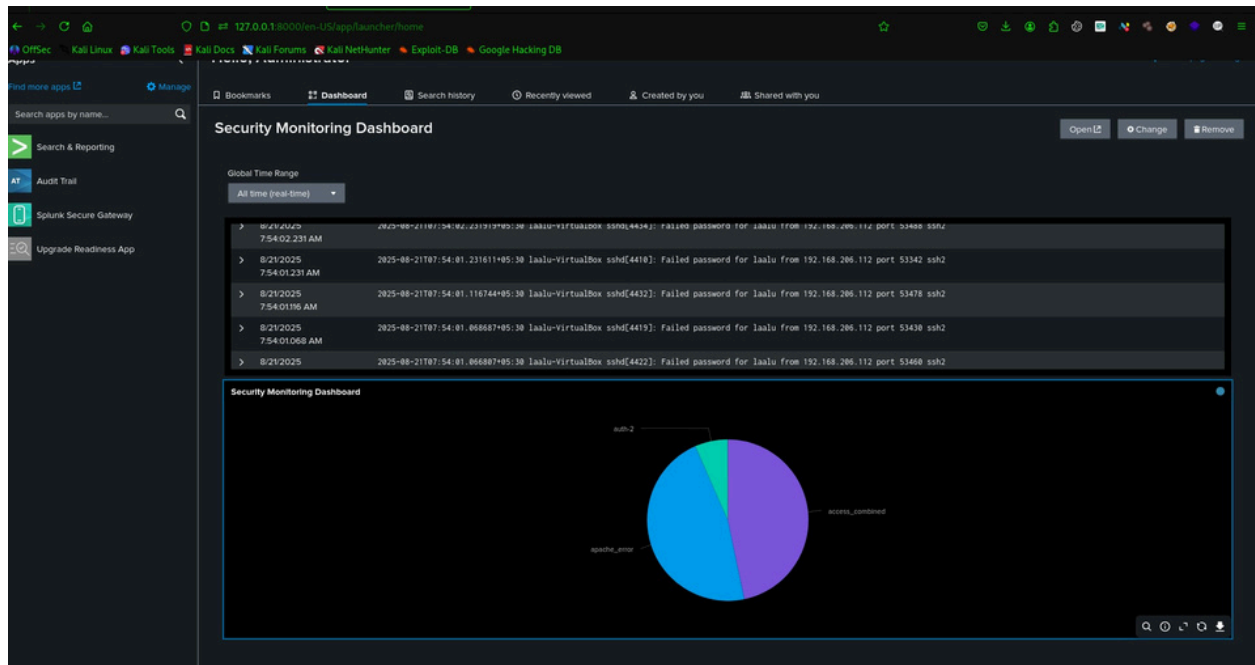
Trigger Condition: Per-Result, Edit

Actions: -1 Action, Add to Triggered Alerts, Edit

Trigger History

20 per page

|    | TriggerTime             | Actions                      |
|----|-------------------------|------------------------------|
| 1  | 2025-08-21 07:59:56 IST | <a href="#">View Results</a> |
| 2  | 2025-08-21 07:59:55 IST | <a href="#">View Results</a> |
| 3  | 2025-08-21 07:59:55 IST | <a href="#">View Results</a> |
| 4  | 2025-08-21 07:59:55 IST | <a href="#">View Results</a> |
| 5  | 2025-08-21 07:59:53 IST | <a href="#">View Results</a> |
| 6  | 2025-08-21 07:59:52 IST | <a href="#">View Results</a> |
| 7  | 2025-08-21 07:59:52 IST | <a href="#">View Results</a> |
| 8  | 2025-08-21 07:59:52 IST | <a href="#">View Results</a> |
| 9  | 2025-08-21 07:59:50 IST | <a href="#">View Results</a> |
| 10 | 2025-08-21 07:59:50 IST | <a href="#">View Results</a> |
| 11 | 2025-08-21 07:59:49 IST | <a href="#">View Results</a> |
| 12 | 2025-08-21 07:59:49 IST | <a href="#">View Results</a> |
| 13 | 2025-08-21 07:59:47 IST | <a href="#">View Results</a> |
| 14 | 2025-08-21 07:59:47 IST | <a href="#">View Results</a> |
| 15 | 2025-08-21 07:59:46 IST | <a href="#">View Results</a> |
| 16 | 2025-08-21 07:59:46 IST | <a href="#">View Results</a> |
| 17 | 2025-08-21 07:59:44 IST | <a href="#">View Results</a> |



Firewall installation and setup (UFW) :

- **UFW:**

- Installed and enabled UFW on the Ubuntu target machine.
- Configured rules to allow essential services (SSH, Splunk Forwarder communication).

```
root@laalu-VirtualBox:/home/laalu# apt install ufw
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ufw is already the newest version (0.36.2-6).
0 upgraded, 0 newly installed, 0 to remove and 276 not upgraded.
root@laalu-VirtualBox:/home/laalu# ufw enable
Firewall is active and enabled on system startup
root@laalu-VirtualBox:/home/laalu# ufw status
Status: active
root@laalu-VirtualBox:/home/laalu#
```

## Blocked ssh bruteforce attack:

The screenshot shows the Splunk Enterprise interface with a search query: `index=* sourcetype="auth-2" "[UFW BLOCK]"`. The search results display 18 events from 8/21/25 12:00:00.000 AM to 8/21/25 10:50:14.000 AM. The events are listed in a table with columns for Time and Event. The events show blocked SSH connections from 192.168.206.112 to 192.168.206.97 on port 22, with the message "UFW BLOCK". The source is identified as /var/log/kern.log and the sourcetype is auth-2.

| Time                    | Event                                                                                                                                                                                                        |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8/21/25 10:48:12.446 AM | 2825-08-21T18:48:12.448888+05:30 IaaS-VirtualBox kernel: [UFW BLOCK] IN=eth0 OUT= MAC=08:00:27:3a:f4:a4:0c:7a:15:fc:e6:f3:08:00 SRC=192.168.206.112 DST=192.168.206.97 LEN=60 TOS=0x00 PREC=0x00 TTL=64 ID=2 |
| 8/21/25 10:48:04.258 AM | 5348 DF PROTO=TCP SPT=56728 DPT=22 WINDOW=64240 RES=0x00 SYN URGP=0 host = IaaS-VirtualBox source = /var/log/kern.log sourcetype = auth-2                                                                    |
| 8/21/25 10:48:00.029 AM | 2825-08-21T18:48:00.029331+05:30 IaaS-VirtualBox kernel: [UFW BLOCK] IN=eth0 OUT= MAC=08:00:27:3a:f4:a4:0c:7a:15:fc:e6:f3:08:00 SRC=192.168.206.112 DST=192.168.206.97 LEN=60 TOS=0x00 PREC=0x00 TTL=64 ID=2 |
| 8/21/25 10:47:58.014 AM | 5349 DF PROTO=TCP SPT=56728 DPT=22 WINDOW=64240 RES=0x00 SYN URGP=0 host = IaaS-VirtualBox source = /var/log/kern.log sourcetype = auth-2                                                                    |
| 8/21/25 10:47:56.989 AM | 2825-08-21T18:47:56.989666+05:30 IaaS-VirtualBox kernel: [UFW BLOCK] IN=eth0 OUT= MAC=08:00:27:3a:f4:a4:0c:7a:15:fc:e6:f3:08:00 SRC=192.168.206.112 DST=192.168.206.97 LEN=60 TOS=0x00 PREC=0x00 TTL=64 ID=2 |
| 8/21/25 10:47:55.969 AM | 5346 DF PROTO=TCP SPT=56728 DPT=22 WINDOW=64240 RES=0x00 SYN URGP=0 host = IaaS-VirtualBox source = /var/log/kern.log sourcetype = auth-2                                                                    |
| 8/21/25 10:47:54.948 AM | 2825-08-21T18:47:54.948873+05:30 IaaS-VirtualBox kernel: [UFW BLOCK] IN=eth0 OUT= MAC=08:00:27:3a:f4:a4:0c:7a:15:fc:e6:f3:08:00 SRC=192.168.206.112 DST=192.168.206.97 LEN=60 TOS=0x00 PREC=0x00 TTL=64 ID=2 |